A MORPHISM OF SQUARES INDUCES A MORPHISM BETWEEN THE SCHEME THEORETIC IMAGES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement displays the two squares; the proof is three lines and consists of the minimality argument using the scheme theoretic inverse image.

Editorial note. The counted spans carry no external references, so no navigation stubs are needed beyond the ones implied by the statement wording.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- The defining property of the scheme theoretic image as the smallest closed subspace through which the morphism factors. Lemma lemma-scheme-theoretic-image (3200-3209), used at proof lines 3375-3378. Gives that Z_1 is contained in the inverse image of Z_2.
- Scheme theoretic inverse images of closed subspaces. used at proof lines 3376-3377. Produces the closed subspace of Y_1 through which f_1 factors.

Lemma 0.2. *Let S be a scheme. Let*

$$\begin{array}{ccc} X_1 & \xrightarrow{f_1} & Y_1 \\ \downarrow & & \downarrow \\ X_2 & \xrightarrow{f_2} & Y_2 \end{array}$$

be a commutative diagram of algebraic spaces over S. Let $Z_i \subset Y_i$, $i = 1, 2$ be the scheme theoretic image of f_i. Then the morphism $Y_1 \to Y_2$ induces a morphism $Z_1 \to Z_2$ and a commutative diagram

$$\begin{array}{ccccc} X_1 & \longrightarrow & Z_1 & \longrightarrow & Y_1 \\ \downarrow & & \downarrow & & \downarrow \\ X_2 & \longrightarrow & Z_2 & \longrightarrow & Y_2 \end{array}$$

Proof. The scheme theoretic inverse image of Z_2 in Y_1 is a closed subspace of Y_1 through which f_1 factors. Hence Z_1 is contained in this. This proves the lemma. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.